Finding the Safety Switches

Inside Small Language Models

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Agenda

Mechanistic interpretability of refusal in small instruct LMs

The question

- Why safety is opaque
- Research question & hypotheses
- Method: patching + ablation

The findings (A–D)

- Sparse & causal
- **Concentrated but not modular**
- Depth $\rightarrow$ modularity
- Generational migration · jailbreaks

Where this goes next

Why look *inside* a safety guardrail?

The motivation

- Instruct LMs are aligned to **refuse** harmful requests — yet jailbreaks still work, and we mostly treat refusal as a *black box*.
- Post-hoc XAI (saliency, attention maps) only *correlates*. **Mechanistic interpretability** reads the actual wiring: *which components causally produce the refusal?*
- If "safety" is a concrete circuit, we can ask the questions that matter:
  - Can we **point to it**? Is it a few heads or smeared across the model?
  - Can we **remove it** cleanly — and what breaks if we do?
  - Does it **look the same** across model families and generations?
- **Scope:** read-only mech-interp — forward passes + hooks (TransformerLens). **No training, no fine-tuning, no gradients.**

Research question & pre-registered hypotheses

What we set out to test

Research question

Are there a **small number of attention heads** that **causally** produce a safety-tuned LM's refusal of harmful prompts — and how does that circuit vary across families and generations?

- **H1 — Sparse:** a handful of heads (≤ 10) explain most refusal.
- **H2 — Causal:** patching those heads *flips* the refusal signal.
- **H3 — Ablation:** zeroing the top-10 drives refusal $\leq 30\%$.
- **H3b — Capability:** … *while preserving capability* (ΔPPL $\leq 5\%$).
- **H4 — Cross-model:** the same structure recurs across models.

Method — localize, then verify causally

9 instruct models · N = 50 matched pairs · single Kaggle T4 · seed 0

Pipeline (per model)

1. **Matched pairs**: harmful (AdvBench) vs benign (HH-RLHF), equal length/topic.
2. **Refusal metric**: refusal-token logit margin (+ regex backstop) at the first generated token.
3. **Activation patching**: swap each head's output benign→harmful; measure Δ(refusal). ⇒ per-head heatmap.
4. **Ablation**: zero / mean-ablate the top-10 heads.
5. **Capability control**: WikiText-2 perplexity — *the key check.*

9 models, 3 families

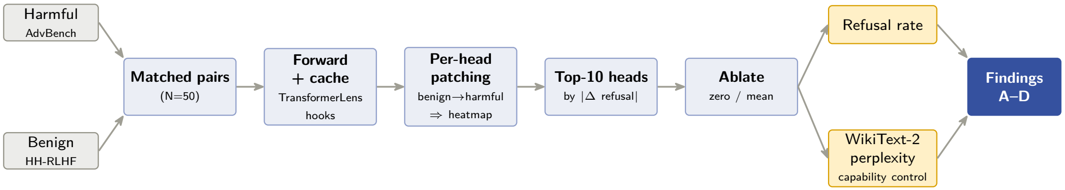
- **Qwen** 1.5 · 2 · 2.5 · 3
- **Gemma** 1 · 2 · 3
- **Llama-3.2** 1B · 3B

Extra probes: K-sweep, last-token & attention-pattern patching, **HarmBench** jailbreak, RealToxicityPrompts continuation.

Within-family generational sweeps let us track how the circuit *moves*.

The pipeline at a glance

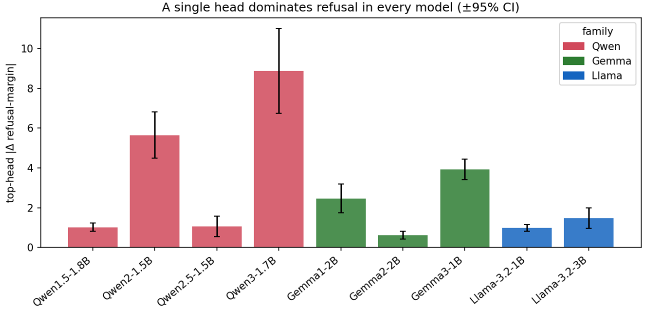
From prompt pairs to causal heads to findings — one model at a time



Frozen model — forward passes + hooks only, **no training**. 9 models, one per session.

Finding: refusal is sparse and causal (H1 ✓ H2 ✓)

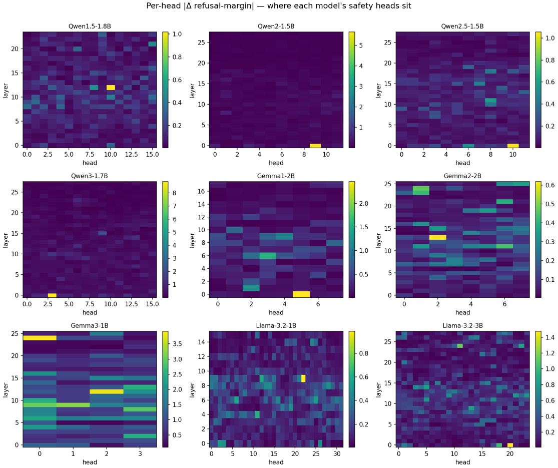
One dominant head carries most of the refusal margin — in all 9 models



- A single head dominates every model, well above its 95% CI: **Qwen3 L0H3 = 8.87 $\pm$ 2.13**.
- Patching one head benign$\rightarrow$harmful *flips* the refusal logit — **causal, 9/9**.
- Heads *cluster*, they are not scattered — but *where* they cluster varies by model (Finding C).

Where the safety heads sit

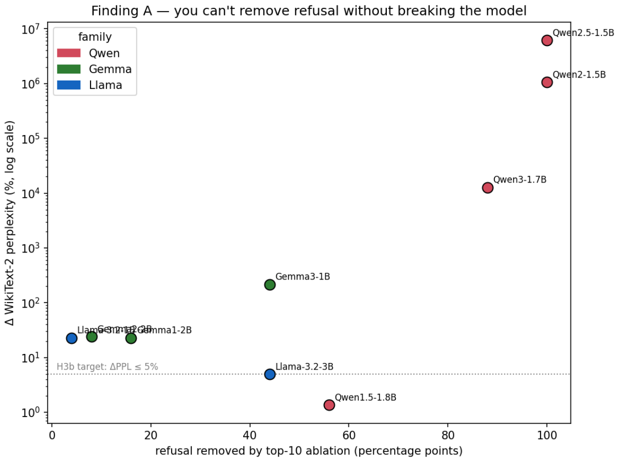
Per-head |Δ refusal-margin| — the patching map for all 9 models



- One panel per model; axes are **layer × head**, colour = causal importance.
- One or two **bright cells** stand out — sparsity, seen directly.
- Their *position* differs across models — the seed of Finding C.

Finding A — concentrated, but not modular

The headline: removing refusal always breaks the model (H3b falsified)



- The 4 models that drop to **0% refusal** are *exactly* the 4 with the biggest perplexity blow-up.
- Qwen2.5: PPL $\times$**61,000**; output is *gibberish*, not compliance.
- No model removes refusal *and* keeps capability $\Rightarrow$ **H3b falsified**.
- Refusal is entangled with the heads the model needs to generate coherent text.

Finding B — modularity scales with circuit *depth*

Where the head sits decides how cleanly you can cut it

Early (layer 0) $\Rightarrow$ catastrophic

- Qwen top head at **L0**: zeroing it destroys coherence (PPL $\times$128 to $\times$61,000).
- These are general-purpose heads that *also* gate refusal.

Late (layer 24) $\Rightarrow$ switch-like

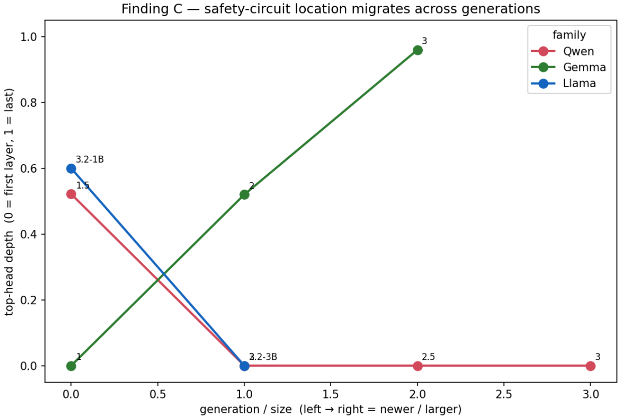
- **Gemma-3-1B L24H0**: refusal $44\% \to 0\%$ at only **+217%** PPL.
- The *only* model where ablation yields genuinely toxic output: **RTP Δtox = +0.128** $(10\times$ any other).

Takeaway

The cleanest, most "switch-like" safety circuit we find is **late-layer**. Early-layer "safety" heads are really load-bearing generation heads.

Finding C — location migrates across generations

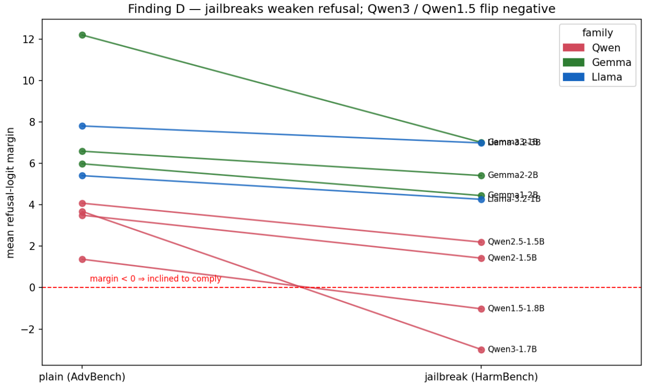
How a family implements refusal is a moving target



- **Gemma** pushes safety *deeper* each generation: L0 → L13 → **L24**.
- **Qwen** pulls it the *other way*: mid-network (L12) → **layer 0** from g2 on.
- **Llama-3B** is a *hybrid*: early (L0) detection *and* late (L24) enforcement.
- A novel within-family result — refusal structure **drifts across releases**.

Finding D — jailbreaks weaken refusal

Robustness is non-monotonic — newer is not safer



- Most robust: **Gemma-2-2B** (96%→96%, no drop); Llama-3B even rises (88%→96%).
- Most brittle: **Qwen3 & Qwen1.5** — refusal margin *flips negative* under HarmBench (inclined to comply).
- **Qwen3** < **Qwen2.5**: the newest model is the most jailbreakable.

Hypothesis scorecard

Sparse & causal hold universally; “clean removal” does not

H	Claim	Verdict
H1	Sparse — a dominant head + short tail	✓ **holds 9/9**
H2	Causal — patching flips the refusal logit	✓ **confirmed 9/9**
H3	Ablation drives refusal $\leq 30\%$	∼ partial (5/9)
H3b	… *and* capability preserved (ΔPPL $\leq 5\%$)	✗ **falsified**
H4	Same structure across models	∼ richer: location *migrates*

- The pre-registered “failure mode” *became* the contribution: refusal is **concentrated but not modular**.

Cross-model results at a glance (N = 50)

Top head · refusal removal · capability cost · jailbreak

Model	L×H	Top head	Refusal clean→abl	ΔPPL	JB refusal	Margin plain→jb
Qwen1.5-1.8B	24×16	L12H10	80%→24%	+1.4%	80→42	$+1.37 \to -\mathbf{1.02}$
Qwen2-1.5B	28×12	L0H9	100%→0%	×10,600	100→74	$+3.50 \to +1.42$
Qwen2.5-1.5B	28×12	L0H10	100%→0%	×61,000	100→94	$+4.08 \to +2.20$
Qwen3-1.7B	28×16	L0H3	88%→0%	×128	88→42	$+3.68 \to -\mathbf{2.99}$
Gemma1-2B	18×8	L0H5	88%→72%	+23%	88→82	$+5.98 \to +4.44$
Gemma2-2B	26×8	L13H2	96%→88%	+24%	**96→96**	$+6.59 \to +5.41$
Gemma3-1B	26×4	L24H0	44%→0%	+217%	44→36	$+12.2 \to +7.0$
Llama-3.2-1B	16×32	L9H22	96%→92%	+23%	96→94	$+5.41 \to +4.26$
Llama-3.2-3B	28×24	L0H20	88%→44%	+5%	88→96	$+7.80 \to +6.99$

Read Finding A straight off the table: the four models at **0% refusal** are exactly the four with the biggest ΔPPL. RTP Δtoxicity is null everywhere except **Gemma-3 (+0.128)**.

Where this goes next

From mapping the circuit to acting on it

Make refusal modular

- Steering vectors / refusal-direction edits & weight surgery instead of blunt ablation.
- **SAEs** to decompose the entangled L0 heads into separable features.
- Test the design lever directly: *can a late-layer circuit be cut cleanly?*

Tighten rigor

- Causal scrubbing; 50-prompt human metric audit; Llama-3B higher-K sweep.

Broaden coverage

- Scale beyond 4B parameters — does the picture hold at larger sizes?
- Multi-axis safety: deception, bias, PII — not just toxicity.
- Cross-lingual refusal circuits.
- More model families, to test how general the depth$\rightarrow$modularity law is.

Conclusion

Safety is mechanistic — but entangled

1. Refusal is **sparse and causal**: one dominant head, in all 9 models (H1/H2 ✓).
2. It is **concentrated but not modular** — removing it always costs capability (H3b ✗). *Refusal-rate alone overstates a clean "safety switch".*
3. **Modularity scales with depth**; the circuit's **location migrates** across generations; **newer $\neq$ safer** under jailbreaks.

Methodological lesson

A **capability control** (perplexity) is mandatory in any ablation study — without it, "refusal $\to$ 0%" reads as success when it is really model breakage.

Thank you — questions welcome.